

Attested Transition Execution for DAZE-Style Simulator Backdoors

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Abstract

Reinforcement-learning (RL) agents are increasingly trained on third-party simulators, making the simulator a supply-chain trust anchor. DAZE shows that a malicious simulator wrapper can implant reward-free backdoors by changing what transition the learner observes after it submits an action without changing the action itself, so action-level logging cannot detect it [1]. Prior RL backdoor work mainly studies policy/agent triggers, competitive backdoors, or sparse targeted attacks [2, 3, 4]; we instead protect the simulator data plane. We study the complementary defense problem under an explicit trust boundary: the simulator core, configuration/assets, random-number state, observation and termination code, reward inputs, and actuator transform are measured and trusted, while the wrapper is untrusted. *Attested Transition Execution* (ATE) accepts a learner-visible update only when an approved measured step relation, exact replay, or an enrolled receipt service vouches for the full transition record. The headline result is zero admitted tampered updates: on original DAZE Hopper-v5 and Reacher-v5 PPO, attack ASR reaches at least 0.8435 and 0.9843, while ATE restores clean-level ASR and matched benign-attested return with ASR-lift 0 and utility 1.0000 over 3.94M defended receipt checks; on SAC Reacher-v5, ASR drops from 0.9996 to 0 with zero forged updates.

Claim and Contributions

Claim. For an approved measured simulator closure with an untrusted wrapper, DAZE-style transition tampering cannot feed forged learner-visible updates into training without failing ATE admission. Positive results support this claim across PPO/SAC continuous-control MuJoCo, Gymnasium classic control, and a custom discrete DQN replay simulator.

Contributions. (i) We frame simulator-wrapper backdoor defense as transition attestation rather than post-hoc anomaly detection. (ii) We show action-only logging is insufficient: a Hopper termination-tamper stress case preserves $a_{\text{sub}} = a_{\text{exec}}$ yet reaches ASR 0.982. (iii) ATE restores original DAZE Hopper/Reacher to clean-level ASR/return, admits zero poisoned updates over 3.94M defended checks, and reduces the action-equality Hopper termination-tamper stress case to ASR 0.009.

Design

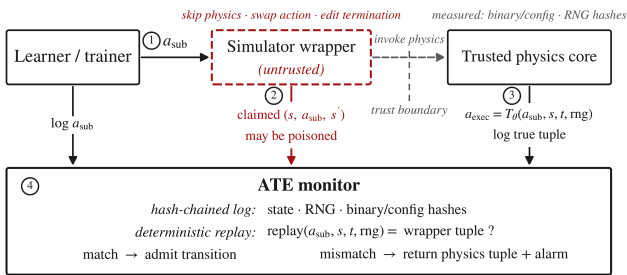


Figure 1. ATE inserts a transition-admission gate between the wrapper and learner.

Figure 1 shows why ATE checks learner-visible transitions rather than wrapper self-reports. The gate admits an update only when a measured step relation or receipt path vouches for actions, state/RNG, and learner-visible fields. ATE admits a transition only if the approved step relation validates this full record:

$$(s, o, a, T_{\theta}(a, s, t, \text{rng}), s', o', r, d, \tau)$$

where s and o denote the pre-step simulator state and learner observation; a denotes the learner-submitted action; $T_{\theta}(a, s, t, \text{rng})$ denotes the certified executed action after the measured actuator transform; s' and o' denote the post-step state and observation; r denotes the reward computed from measured inputs; d denotes termination/truncation; and τ denotes receipt metadata such as nonce, sequence, lane, and key epoch. T_{θ} covers legitimate clipping, scaling, action repeat, torque saturation, or safety projection. Exact mode replays deterministic CPU simulators. Receipt mode follows measured-evidence principles from remote attestation architectures [5] and checks signatures, freshness, lane identity, and hash-chain continuity.

Positive Learning Evidence

Setting	Learner	Clean	Attack	ATE	Utility	Admit
Hopper-v5	PPO	0.0177	0.8435	0.0177	1.0000	0
DAZE Reacher-v5	PPO	0.0000	0.9843	0.0000	1.0000	0
DAZE Reacher-v5	SAC	0.0000	0.9996	0.0000	clean-level	0
DAZE CartPole record	DQN	0.415	0.961	0.415	clean-level	0/80,350
tamper Discrete replay relabel	DQN	0.000	0.908/0.933	0.000	0.941	0/all

Table 1. Positive learning evidence across MuJoCo, Gymnasium, and custom replay settings.

Table 1 reports attack success rate (ASR) for clean, unfiltered attack, and ATE-defended training. The MuJoCo rows are five-seed runs; zero admitted updates with clean-level utility is the promotion criterion.

Systems Evidence

Audit or benchmark	Rows/records	Failures
MuJoCo promotion validation	168	0
Software-root protocol v3	57	0
AMC-ATE v2 stress	71	0
Brax/JAX GPU receipts	1.57M	0

Table 2. Systems evidence for the ATE admission path.

Table 2 counts validation checks or receipt records; failures are accepted violations or false rejects. The audits reject key/closure drift, replay, lane mismatch, transition edits, and unsafe transforms; code and result tables are available in our public repository [6].

Boundary and Discussion Prompt

The positive claim is intentionally scoped: the approved closure includes the simulator core, configuration/assets, RNG schedule, observation and termination code, reward inputs, and allowed actuator transforms. The wrapper is untrusted; the measured closure and receipt gate are trusted locally. Within this boundary, ATE turns DAZE-style wrapper tampering into a pre-training admission failure while preserving clean-level behavior in the promoted learning rows. **Prompt:** what root should bind receipt keys for remote batched RL-TPM/TEE quotes, signed step services, or scheduler-enforced measured jobs?

References

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